

Unit 4, Lesson 7: Finding Radius and Diameter

Benchmark

MA.7.GR.1.3 Explore the proportional relationship between circumferences and diameters of circles. Apply a formula for the circumference of a circle to solve mathematical and real-world problems.

Learning Target

- ☐ I can find the radius and diameter of circles, given the circumference.

4.7.1 Warm-Up: Everglades National Park

The Florida Everglades is home to two native species that are considered to be close to extinction. Cape Sable seaside sparrows and the Florida leafwing butterflies live only within Everglades National Park and the immediately adjacent lands.

Cape Sable seaside sparrows are small birds about 13 centimeters long.

The Florida leafwing is a medium-sized butterfly approximately 77 millimeters in length.



- 1. Which species has the largest length?

Andrew multiplies the sparrow’s length by 0.4 to estimate the length of the sparrow in inches. To estimate the length of the butterfly in inches he multiplies the length by 0.04.

- 2. Use Andrew’s method to determine the length of the sparrow and the length of the butterfly in inches.

Cape Sable Seaside Sparrow Length (inches)	Florida Leafwing Length (inches)

4.7.2 Exploration: Using Circumference to Find Other Measurements

1. A circular tank at a wastewater treatment facility in Davie, Florida has walkways built around the edge of the tank and across the center of the tank so that water management workers can check and treat the water. The walkway built around the top of the tank, shown in pink, is 37.68 meters (m) long.



Discuss with your partner how this information could be used to find the distance of the walkway that runs across the top of the water tank. Write a summary of your discussion.

2. Josephine said the formula for the circumference of a circle could be used to find the diameter. Her partial explanation and work are shown. Work with your partner to complete Josephine’s work and explanation by filling in the boxes.

Josephine’s Work	Explanation
$C = \pi d$	Begin with the formula for circumference of a circle in terms of diameter.
$= 3.14 \cdot d$	Substitute _____ for the value of the circumference (C) and 3.14 for pi.
$\frac{3.14}{3.14} = \frac{3.14}{3.14} \cdot d$	Isolate the diameter (d) by dividing both sides of the equation by _____.
$= d$	Calculate the diameter.

3. Some tanks at wastewater treatment centers have walkways built only to the center of the tank.
- a. What part of a circle does this represent?



- b. What would be the length of a walkway like this in Davie, FL?

4.7.3 Guided Practice: Using Circumference to Find Other Measurements

1. Pizzeria and Little Italy are both offering large pizzas for \$12.99 each. Victor knows that Pizzeria's pizza has a circumference of 43.96 inches (in.) and Little Italy's pizza has a diameter of 13.25 in. Which pizza has the largest diameter?

2. Fredrick is a landscape architect and is updating a circular garden. He plans to put a fountain in the center of the garden but does not know the garden's radius.

If the garden has a circumference of 54.95 feet (ft), what is the radius?

3. Mary recently baked both an apple and cherry pie. The apple pie has a diameter of $23\frac{1}{2}$ centimeters (cm), and the cherry pie has a circumference of 71.78 cm.

- a. What is the radius of the cherry pie?

- b. Which pie has the larger radius?

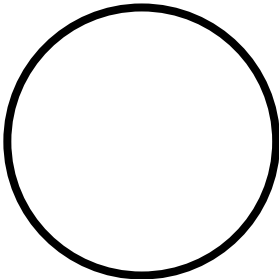
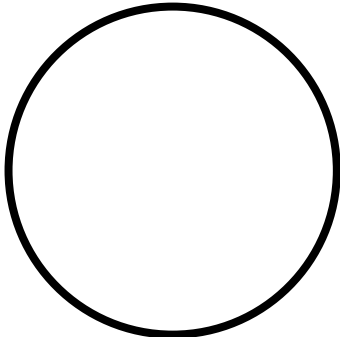
4.7.4 Your Turn!

1. Fort Lauderdale recently installed a new traffic roundabout. At its widest part, the roundabout has a circumference of 345.4 feet (ft). What is the radius of the traffic roundabout at its widest part?
2. The widest part of a super bouncer ball has a circumference of $\frac{17\pi}{20}$ decimeters (dm).
 - a. What is the diameter of the super bouncer ball?
 - b. If the widest part of a regular bouncy ball has a circumference of 0.49π dm, what is the diameter?
 - c. What is the difference in diameter of the super bouncer and the regular bouncy ball?
3. The High Roller in Las Vegas, Nevada is the world's largest Ferris wheel. The High Roller has a circumference of 158.5π m. At the highest point on the Ferris wheel, how far from the center is the rider?



4.7.5 Wrap-Up: Ticket Out the Door

1. Find the diameter and radius for circles with the given circumferences. Use $\pi = 3.14$. If necessary, round your answers to the nearest thousandth.

Circle	 Circumference $\frac{5}{2}$ meters (m)	 Circumference 10.99 inches (in.)
Diameter		
Radius		

